

Robotics Rover — Cheat Sheet

Chain K — Hardware & Systems Foundations · one-page recall, not first-time learning

Motor Control

Term	Meaning
H-bridge	Lets a low-current logic signal control a high-current motor's direction
PWM	Duty cycle sets the effective average voltage → speed
Best default driver	TB6612FNG over L298N — lower voltage drop, runs cooler
Size for	Stall current, not typical running current

Power Isolation

Rule	Detail
Two power domains	Motor battery + logic supply, meeting ONLY at ground
Brownout cause	Shared rail sags below the microcontroller's minimum voltage
Flyback diode	Protects driver from motor back-EMF spikes

Chassis & Sensing

Fact	Detail
Gearbox choice	Metal holds up under load; plastic can strip over time
IR line sensor	Measures reflected IR — light background reflects more than a dark line
Sensor mount height	~5-10mm above the surface

Control Loops

Behavior	Cause
P-only control	Tends to overshoot/oscillate
P+D control	Derivative term damps oscillation, converges smoothly
Growing oscillation	Insufficient damping (low Kd relative to Kp) — increase Kd, not just lower Kp
Outright divergence	Kp too high even with damping present